

Scalar-Carrier Recoverability-Gradient Probe Release for HAOS-IIP

Tomislav RupiĆ
Independent Researcher

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Abstract

This paper freezes release 52.1 of HAOS-IIP as the first bounded recoverability-gradient probe on the validated scalar geometry carrier. The probe uses the same mean-zero scalar Green field as a deterministic recoverability potential and the same bounded local metric field established by 51.4. The resulting response proxy is $J_{\text{rec}} = -G\nabla\phi$. The result is mixed and therefore boundedly open. Across the tested mild-disorder and bounded kernel-family window, the response field is already strongly radial, but its shell-law closure is not yet positive: the inferred power-law slopes do not stabilize near -2 , the shell-profile fit quality remains weak, and the scaled-flux profiles do not remain constant enough across refinement. The correct conclusion is therefore not a force-law claim, but a cleaner boundary: the scalar carrier and local metric field are now in place, while the first response-law reconstruction is not yet strong enough to support a positive inverse-square-like recoverability-gradient statement.

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1 Scope

Release 52.1 is the first response-style probe on the scalar carrier.

It does not add:

- a universal force law;
- a fundamental ontic force claim;

- closure on arbitrary substrates or strong disorder;
- current or curvature closure.

It does add:

- the first deterministic operator-native recoverability-gradient proxy on the scalar carrier;
- a precise answer about what already works and what still does not;
- a sharper next-step target for response reconstruction.

2 Probe Definition

The probe stays entirely inside the existing scalar carrier:

- recoverability potential: the same mean-zero scalar Green field ϕ ;
- local geometry: the bounded coarse local metric field G from 51.4;
- effective response proxy:

$$J_{\text{rec}} = -G\nabla\phi.$$

This is intentionally weaker than a force claim. It is only the first bounded response proxy on the same validated carrier.

3 What Was Tested

The bounded test asked whether the resulting response field is:

- strongly radial;
- close to inverse-square in shell profile;
- close to constant in $r^2 F_r$;
- stable under the same mild-disorder and bounded kernel-family window.

4 Frozen Quantitative Summary

5 What 52.1 Now Supports

The strongest honest statement now supported by the repository is:

on the validated scalar carrier, the Green recoverability potential already supports a coherent radial response field, but the first inverse-square-like response-law closure is still open.

That is not a null result in the empty sense. It cleanly separates:

- carrier closure and local metric closure, which are already positive;
- response-law closure, which is not yet positive.

Channel	Status	Frozen read
Radial alignment	positive	All tested cases remain strongly radial with alignment around 0.99+.
Power-law closure	open	Clean slopes range from about -0.90 at $n = 11$ to about -1.28 at $n = 13$, not a stable -2 family.
Fit quality	open	Shell-profile fit R^2 remains too weak.
Scaled-flux constancy	open	$r^2 F_r$ remains too variable across shells and across refinement.
Refinement drift	profile open	Maximum profile drift reaches 0.5249.

Table 1: Bounded recoverability-gradient summary frozen by release 52.1.

6 Why This Is Still Progress

The important point is what did *not* fail:

- the scalar geometry carrier itself;
- the bounded local metric field;
- the directional coherence of the response field.

What failed is the first law reconstruction. That means the next step should improve the response estimator, not reopen the carrier.

7 The Right Next Move

The correct follow-up is 52.2:

- improve the response reconstruction itself;
- test shell-native or law-aware gradient extraction;
- only then ask again whether one inverse-square family closes.

8 Conclusion

Release 52.1 freezes the first bounded recoverability-gradient probe on the scalar geometry carrier.

The correct read is disciplined: a force-like response statement is not yet earned. But the repository now has a sharper and more useful boundary than before. The carrier exists, the local metric field exists, the response direction is already coherent, and the first response-law closure is now isolated as its own concrete open problem.